Spatial Estimates of SWE for the Rio Grande Basin

April 2, 2025

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Summary of SNODAS SWE for April 1, 2025

Spatial estimates of snow water equivalent (SWE; presented in inches) for the Rio Grande (RG) Basin study area are presented in Figure 1. Estimated SWE (presented in median and mean depths in inches (in) and volumes in acre-feet (af)) is reported for the standard Hydrologic Unit Code 6 (HUC6) basins (Rio Grande Headwaters, Upper Rio Grande, and Rio Grande-Elephant Butte; Table 1) and HUC8 subbasin groupings (Upper Rio Grande, Rio Chama, Sangre de Cristo, and Jemez; Table 2). Estimated SWE (presented in median and mean depths (in) and volumes (af)) by basin and elevation band are reported in Table 3.

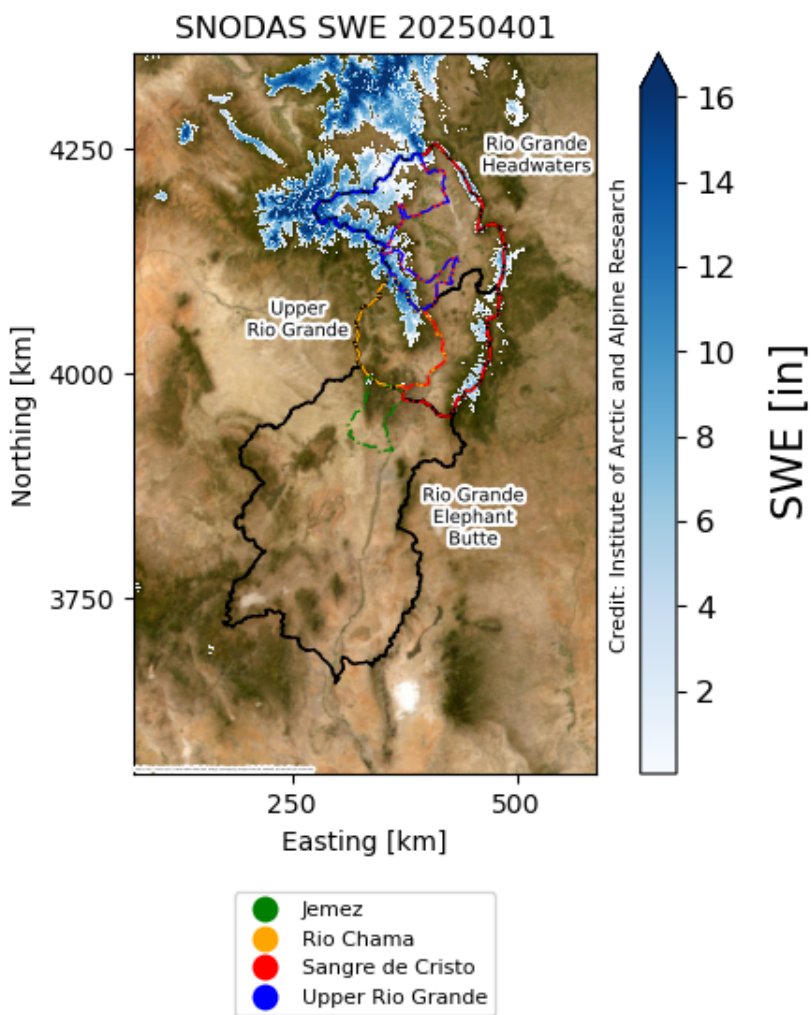

Figure 1. Estimated SNODAS SWE (in.) across the Rio Grande Basin study area for April 1, 2025.

Introduction

The Rio Grande (RG) is a snowmelt-driven river system with up to 75% of its water originating from seasonal snowpack in the mountains of Colorado and northern New Mexico (Moeser et al., 2021; Figure 2). Snowpack acts as a natural reservoir in the RG Basin, gradually releasing meltwater during the spring and early summer months, sustaining streamflow and providing vital water supply for downstream users. However, regional decreases in snowpack volume and earlier snowmelt timing, especially throughout the western U.S. and New Mexico, can lead to reduced water availability (Mote et al., 2005; EPA, 2016b; Mote et al., 2018; Hale et al., 2023). Long-term changes in snowpack patterns and timing can serve as an indicator of climate change, while understanding short-term changes is critical for monitoring seasonal variability and forecasting seasonal water supply. Monitoring snowpack provides valuable data on shifts in precipitation patterns and temperatures, helping scientists and decision-makers better understand and manage climate-related challenges.

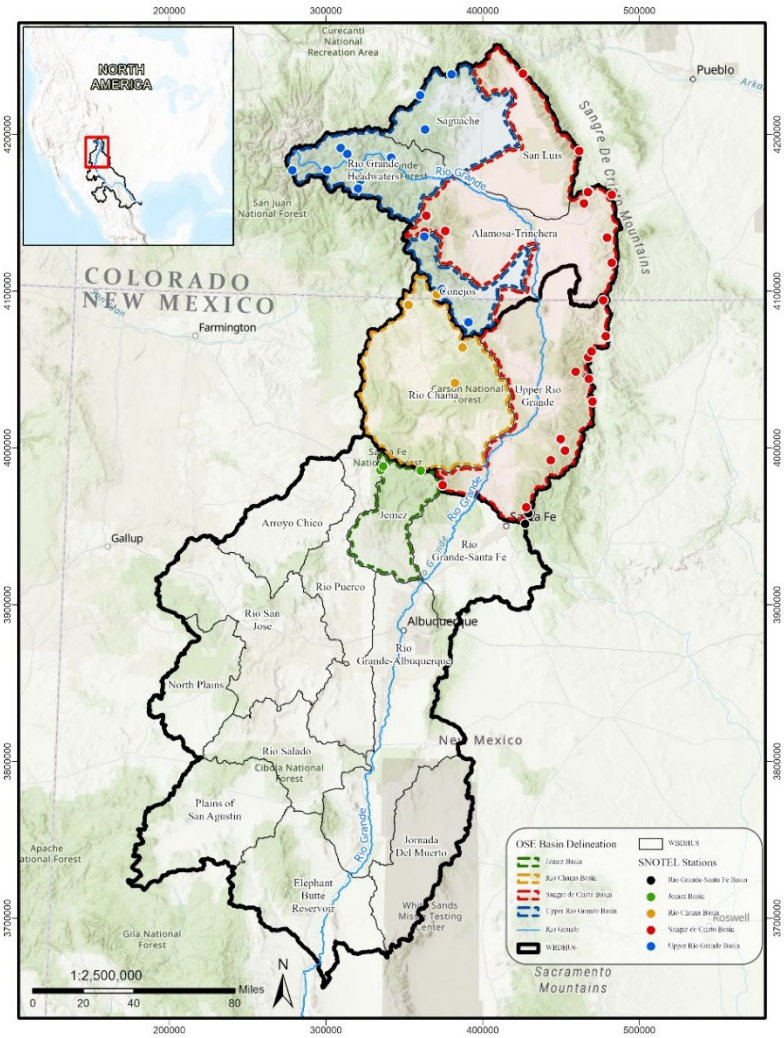

Figure 2. Overview map of the Rio Grande (RG) Basin study area. *The Rio Grande Headwaters, Upper Rio Grande, and Rio Grande-Elephant Butte Basins are presented as the standard HUC6 basins. HUC8 groupings (Jemez – green, Rio Chama – orange, Sangre de Cristo – red, and Upper Rio Grande – blue) are denoted.*

Snow water equivalent (SWE) is the estimate of the depth of liquid water contained in the snowpack. SWE is the standard metric in estimating the volume of meltwater in the snowpack, which plays a critical role in the seasonal snow water supply in the RG Basin. SWE is measured at in-situ Snow Telemetry (SNOTEL) sites throughout the RG Basin and the greater Western U.S. However, the spatial distribution of SWE is complex and often difficult to estimate outside of the in-situ SNOTEL measurements. While SNOTEL provides important point-based measurements, it has been suggested that these data may not be representative of the surrounding snow-covered area (Meromy et al., 2013). Whereas regional tendencies of spatial distribution are consistent, local snowpack volumes are more variable, all of which influences the magnitude and timing of snowmelt-driven runoff.

SWE Analysis Tool

The RG Basin, and more broadly the Western U.S., is experiencing a prolonged and severe drought (EPA, 2016a), exacerbated by a warming climate that increases evapotranspiration, reduces overall snowpack quantity, and reduces the seasonal duration of snowpack. The application of the SWE Analysis Tool (SWEAT) will supply scientists and decision-makers with an improved insight into the spatial distribution of SWE by providing real-time spatial SWE estimates for the RG Basin, its subbasins, and elevation bands within the basin.

Currently, SWEAT incorporates the SNODAS gridded SWE data product (NOHRSC, 2004). Because no spaceborne sensor can directly measure SWE, spatial distribution of SWE must be estimated from remotely sensed observations coupled with modeling approaches. However, gridded SWE products, like SNODAS, are known to have accuracy issues across both mountainous and non-mountainous regions (Mudryk et al., 2025), resulting in a range of spatial SWE estimates for any given location. NM OSE and INSTAAR are working to integrate more data products to build SWEAT based on an ensemble of spatially estimated SWE. By taking an ensemble approach, the goal for SWEAT is to provide spatial SWE estimations for the RG Basin that are, on average, equally or more accurate than any individual model (Thompson, 1977; Arsenault et al., 2015). In addition, the use of a single estimate calculated from an ensemble can reduce confusion for water managers about which product may be performing best, providing a path towards consistency throughout users, and is useful in identifying product outliers.

SNODAS

The SNOw Data Assimilation System (SNODAS) represents a modeling and data assimilation system developed by the National Operational Hydrologic Remote Sensing Center (NOHRSC). SNODAS generates spatial estimates of SWE, snow cover, and snow depth through a combination of remotely sensed snow data and in-situ measurements of SWE from SNOTEL at a 1-km spatial resolution and 24-hour temporal resolution.

The national SNODAS dataset is available through the SNODAS Data Products at National Snow and Ice Data Center (NSIDC), Version 1 (NOHRSC, 2004). SNODAS data can be viewed at the NOHRSC Interactive Snow Information page. Gridded observed snowfall images, seasonal totals, and data downloads in several formats can be found on the National Gridded Snowfall Analysis page.

Refer to the SNODAS User Guide and the NSIDC Special Report 11 (Barrett, 2003) for more information.

Discussion

Figures and tables for the identified basins (Figure 1) are presented below. The median and mean SWE (in.) are presented to avoid potentially skewed outliers, most notably due to an uneven distribution of SWE across elevation bands. While the mean value represents the average value, it is highly influenced by outliers and is typically best used for uniform distributions of data. In contrast, the median value represents the middle value and is less sensitive to outliers, which can be more useful in skewed distributions. Note that the basin-wide median and mean values may not be directly comparable with the SNOTEL basin-wide values. If comparisons are made with SNOTEL sites, a better comparison may be made across elevation bands (Table 3).

In general, the spatial distribution of snowpack and SWE volumes are greater across the western subbasins of the Rio Grande Headwaters and Upper Rio Grande (Figures 3 and 4; Table 1), also identified as the Upper Rio Grande subbasin grouping (Table 2). Overall, the eastern subbasins (Figures 3 and 4; Table 1), also identified as the Sangre de Cristo subbasin grouping (Table 2), contain less snowpack and lower SWE volumes.

Rio Grande Headwaters

- Much of the spatial distribution of SWE across the Basin is concentrated within the Rio Grande Headwaters Subbasin (405,555 af – 55% of the estimated basin SWE volume; Table 1) and at elevations greater than 10,000 ft. (Table 3).

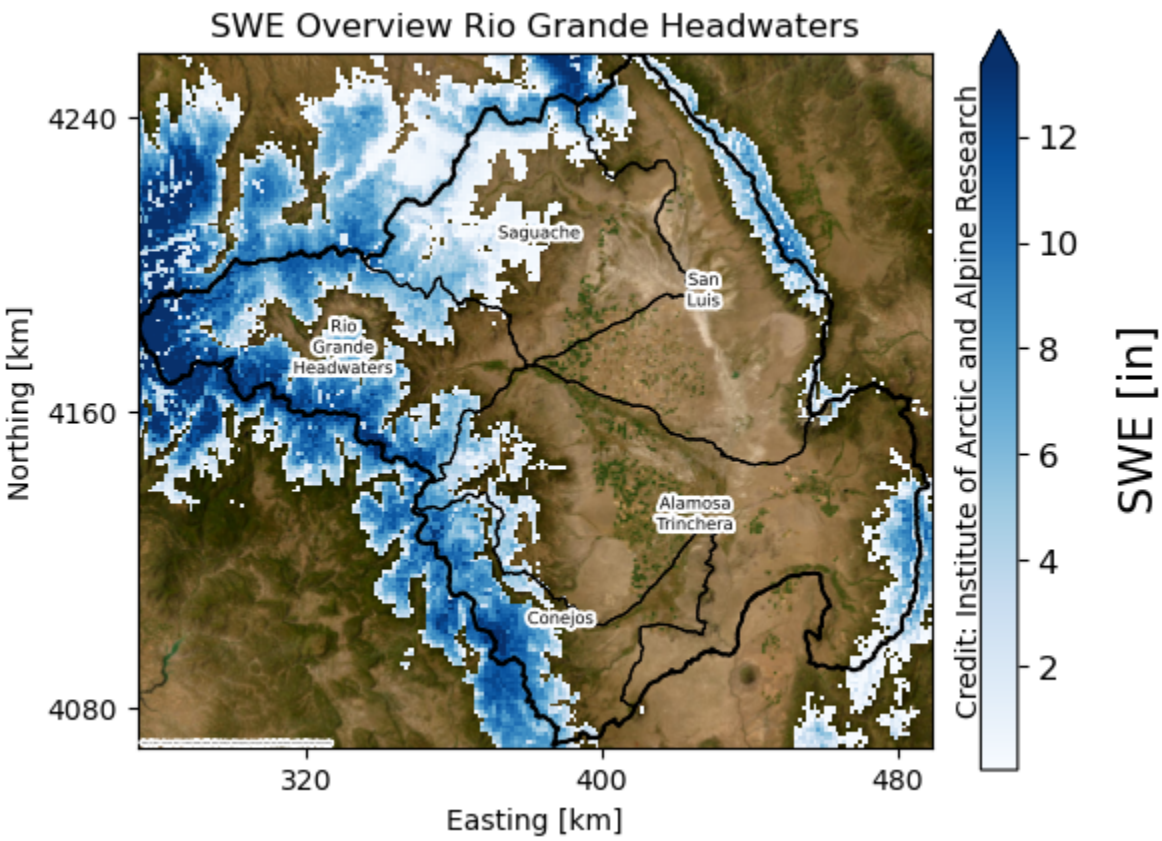

Figure 3. Estimated SNODAS SWE (in.) across the Rio Grande Headwaters Basin for April 1, 2025. *SWE amount (in.) for the Rio Grande Headwaters Basin; including the Rio Grande Headwaters, Alamosa-Trinchera, San Luis, Saguache, and Conejos subbasins.*

Upper Rio Grande

- Much of the spatial distribution of SWE across the Basin is concentrated within the Rio Chama Subbasin (148,912 af – 66% of the estimated basin SWE volume; Table 1) and at elevations greater than 9,000 ft. (Table 3).

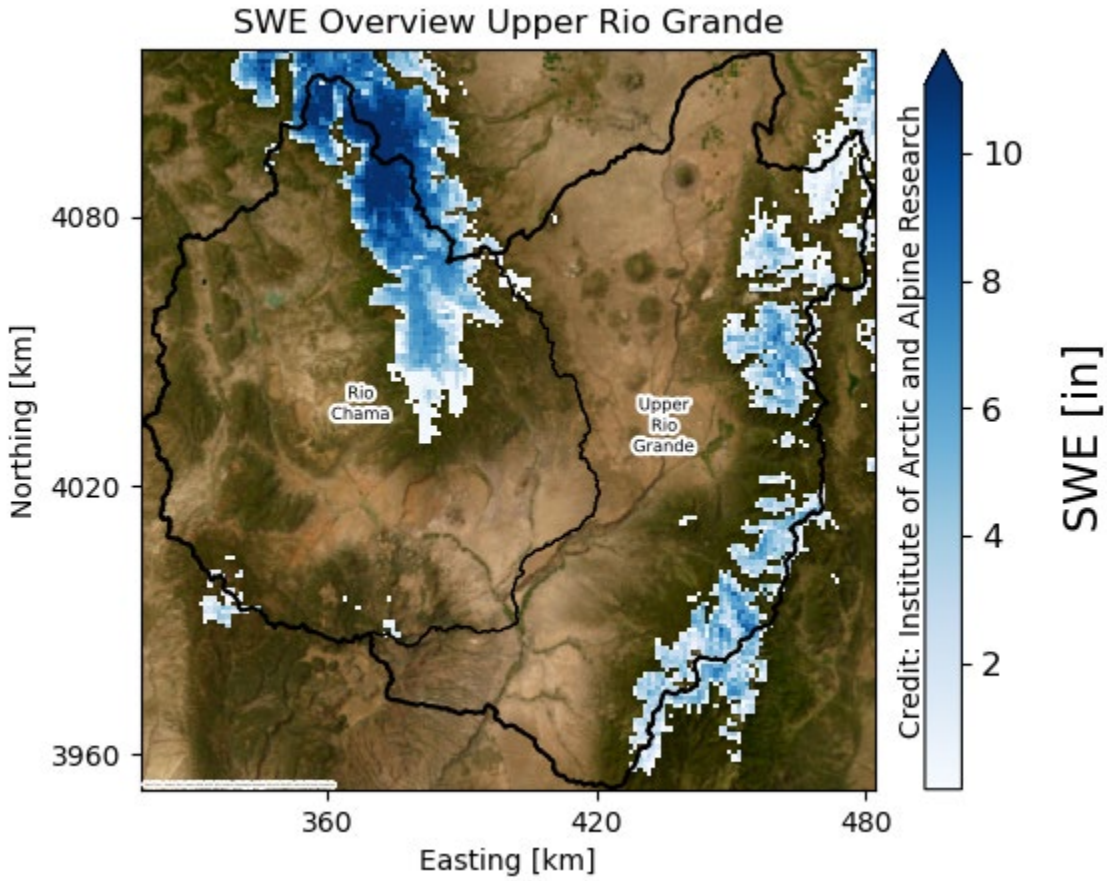

Figure 4. Estimated SNODAS SWE (in.) across the Upper Rio Grande Basin for April 1, 2025. *SWE amount (in.) for the Upper Rio Grande Basin; including the Upper Rio Grande and Rio Chama subbasins.*

Rio Grande-Elephant Butte

- Currently, three subbasins (Rio Grande-Santa Fe, Jemez, and Rio Puerco) contribute to the SWE estimations across the Basin (Table 1).
- Much of the spatial distribution of SWE across the Basin is concentrated within Jemez Subbasin (1,601 af – 72% of the estimated basin SWE volume; Table 1) and at elevations greater than 9,000 ft. (Table 3).

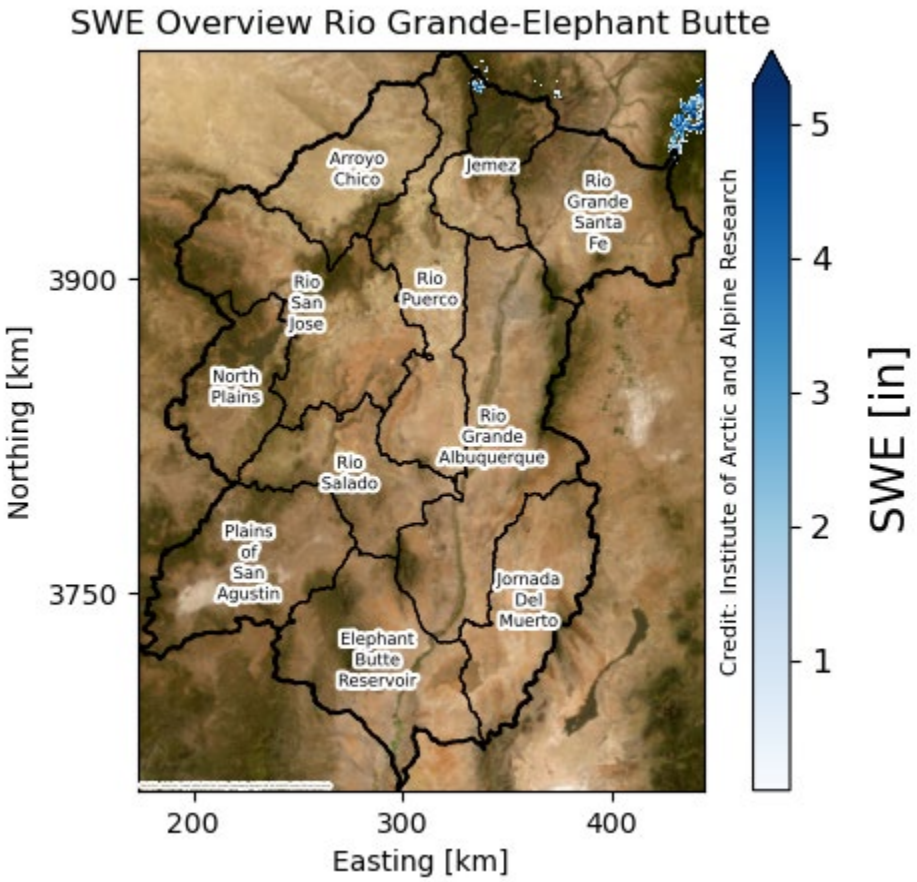

Figure 5. Estimated SNODAS SWE (in.) across the Rio Grande-Elephant Butte Basin for April 1, 2025. *SWE amount (in.) for the Rio Grande-Elephant Butte Basin; including the Rio Grande-Santa Fe, Jemez, Rio Grande-Albuquerque, Arroyo Chico, North Plains, Rio San Jose, Plains of San Agustin, Rio Salado, Jornada Del Muerto, Elephant Butte Reservoir, and Rio Puerco subbasins.*

Tables

Table 1. Estimated SWE by HUC6 Basin for April 1, 2025. *Shown are median SWE (in.), mean SWE (in.), and estimated volume (af) for the current report date. Values are separated by HUC8 subbasin and totaled for each HUC8 basin.*

HUC6 Basin	HUC8 Subbasin	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)
Rio Grande Headwaters	Rio Grande Headwaters	5.4	5.5	405,555
	Alamosa-Trinchera	0.0	0.5	68,916
	San Luis	0.0	0.5	42,448
	Saguache	0.0	1.1	80,492
	Conejos	0.0	3.5	141,290
	Total	**0.0**	**1.8**	**738,702**
Upper Rio Grande	Upper Rio Grande	0.0	0.4	75,207
	Rio Chama	0.0	0.9	148,912
	Total	**0.0**	**0.7**	**224,118**
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	0.0	0.0	420
	Jemez	0.0	0.0	1,601
	Rio Grande-Albuquerque	0.0	0.0	0
	Arroyo Chico	0.0	0.0	0
	North Plains	0.0	0.0	0
	Rio San Jose	0.0	0.0	0
	Plains of San Agustin	0.0	0.0	0
	Rio Salado	0.0	0.0	0
	Jornada Del Muerto	0.0	0.0	0
	Elephant Butte Reservoir	0.0	0.0	0
	Rio Puerco	0.0	0.0	195
	Total	**0.0**	**0.0**	**2,216**

Table 2. Estimated SWE by Specified Subbasin Groupings for April 1, 2025. *Shown are median SWE (in.), mean SWE (in.), and estimated volume (af) for the current report date. Values are separated by HUC8 subbasin and totaled for each specified subbasin grouping.*

Subbasin Grouping	HUC8 Subbasin	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)
Upper Rio Grande	Rio Grande Headwaters	5.4	5.5	405,555
	Saguache	0.0	1.1	80,492
	Conejos	0.0	3.5	141,290
	Total	**0.4**	**3.4**	**625,955**
Sangre de Cristo	San Luis	0.0	0.5	42,448
	Alamosa-Trinchera	0.0	0.5	68,916
	Upper Rio Grande	0.0	0.4	75,207
	Total	**0.0**	**0.5**	**185,471**
Rio Chama		**0.0**	**0.9**	**149,906**
Jemez		**0.0**	**0.0**	**1,601**

Table 3. Estimated SWE by HUC6 Basin and Elevation Band for April 1, 2025. *Shown are median SWE (in.), mean SWE (in.), and estimated volume (af) for the current report date. Values are separated by elevation band (1000 ft. increments) and HUC8 subbasin.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)
Rio Grande Headwaters	Rio Grande Headwaters	7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	0
		9000-10000'	0.0	0.7	9,865
		10000-11000'	5.6	5.7	125,119
		11000-12000'	9.5	9.4	205,773
		12000-13000'	11.2	10.7	63,896
		13000-14000'	9.6	7.3	902
	Alamosa-Trinchera	7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	0
		9000-10000'	0.0	0.1	2,300
		10000-11000'	1.6	2.4	28,826
		11000-12000'	3.5	3.9	33,621
		12000-13000'	1.2	2.2	4,136
		13000-14000'	0.4	0.4	32
	San Luis	7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	0
		9000-10000'	0.0	0.2	1,451
		10000-11000'	1.1	2.3	13,904
		11000-12000'	3.5	3.9	18,355
		12000-13000'	2.3	3.4	8,700
		13000-14000'	0.0	0.6	37
	Saguache	7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	64
		9000-10000'	0.3	0.5	8,754
		10000-11000'	3.0	3.1	38,151
		11000-12000'	6.1	6.0	28,056
		12000-13000'	5.6	5.9	5,443
		13000-14000'	1.2	1.2	24
	Conejos	7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	51
		9000-10000'	0.0	2.2	23,513
		10000-11000'	8.7	8.2	77,310
		11000-12000'	8.4	7.8	38,153
		12000-13000'	5.8	5.8	2,263

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)
Upper Rio Grande	Upper Rio Grande	5000-6000'	0.0	0.0	0
		6000-7000'	0.0	0.0	0
		7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	149
		9000-10000'	0.0	0.4	9,252
		10000-11000'	1.9	2.3	37,924
		11000-12000'	3.3	3.3	26,167
		12000-13000'	0.8	1.5	1,715
	Rio Chama	5000-6000'	0.0	0.0	0
		6000-7000'	0.0	0.0	0
		7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	1,120
		9000-10000'	0.6	2.4	49,794
		10000-11000'	8.4	8.1	90,199
		11000-12000'	9.6	9.3	7,469
		12000-13000'	8.0	8.0	330
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	5000-6000'	0.0	0.0	0
		6000-7000'	0.0	0.0	0
		7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	0
		9000-10000'	0.0	0.0	0
		10000-11000'	1.4	1.4	353
		11000-12000'	0.6	0.8	67
	Jemez	5000-6000'	0.0	0.0	0
		6000-7000'	0.0	0.0	0
		7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	0
		9000-10000'	0.0	0.2	1,032
		10000-11000'	0.0	0.6	569
	Rio Grande-Albuquerque	4000-5000'	0.0	0.0	0
		5000-6000'	0.0	0.0	0
		6000-7000'	0.0	0.0	0
		7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	0
		9000-10000'	0.0	0.0	0
		10000-11000'	0.0	0.0	0
	Arroyo Chico	5000-6000'	0.0	0.0	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)
Rio Grande-Elephant Butte	Arroyo Chico	6000-7000'	0.0	0.0	0
		7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	0
		9000-10000'	0.0	0.0	0
		10000-11000'	0.0	0.0	0
	North Plains	6000-7000'	0.0	0.0	0
		7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	0
	Rio San Jose	5000-6000'	0.0	0.0	0
		6000-7000'	0.0	0.0	0
		7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	0
		9000-10000'	0.0	0.0	0
		10000-11000'	0.0	0.0	0
	Plains of San Agustin	6000-7000'	0.0	0.0	0
		7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	0
		9000-10000'	0.0	0.0	0
	Rio Salado	4000-5000'	0.0	0.0	0
		5000-6000'	0.0	0.0	0
		6000-7000'	0.0	0.0	0
		7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	0
		9000-10000'	0.0	0.0	0
	Jornada Del Muerto	4000-5000'	0.0	0.0	0
		5000-6000'	0.0	0.0	0
		6000-7000'	0.0	0.0	0
		7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	0
	Elephant Butte Reservoir	4000-5000'	0.0	0.0	0
		5000-6000'	0.0	0.0	0
		6000-7000'	0.0	0.0	0
		7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	0
		9000-10000'	0.0	0.0	0
		10000-11000'	0.0	0.0	0
	Rio Puerco	4000-5000'	0.0	0.0	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)
Rio Grande-Elephant Butte	Rio Puerco	5000-6000'	0.0	0.0	0
		6000-7000'	0.0	0.0	0
		7000-8000'	0.0	0.0	0
		8000-9000'	0.0	0.0	0
		9000-10000'	0.0	0.1	60
		10000-11000'	0.0	0.3	135

Acknowledgments

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